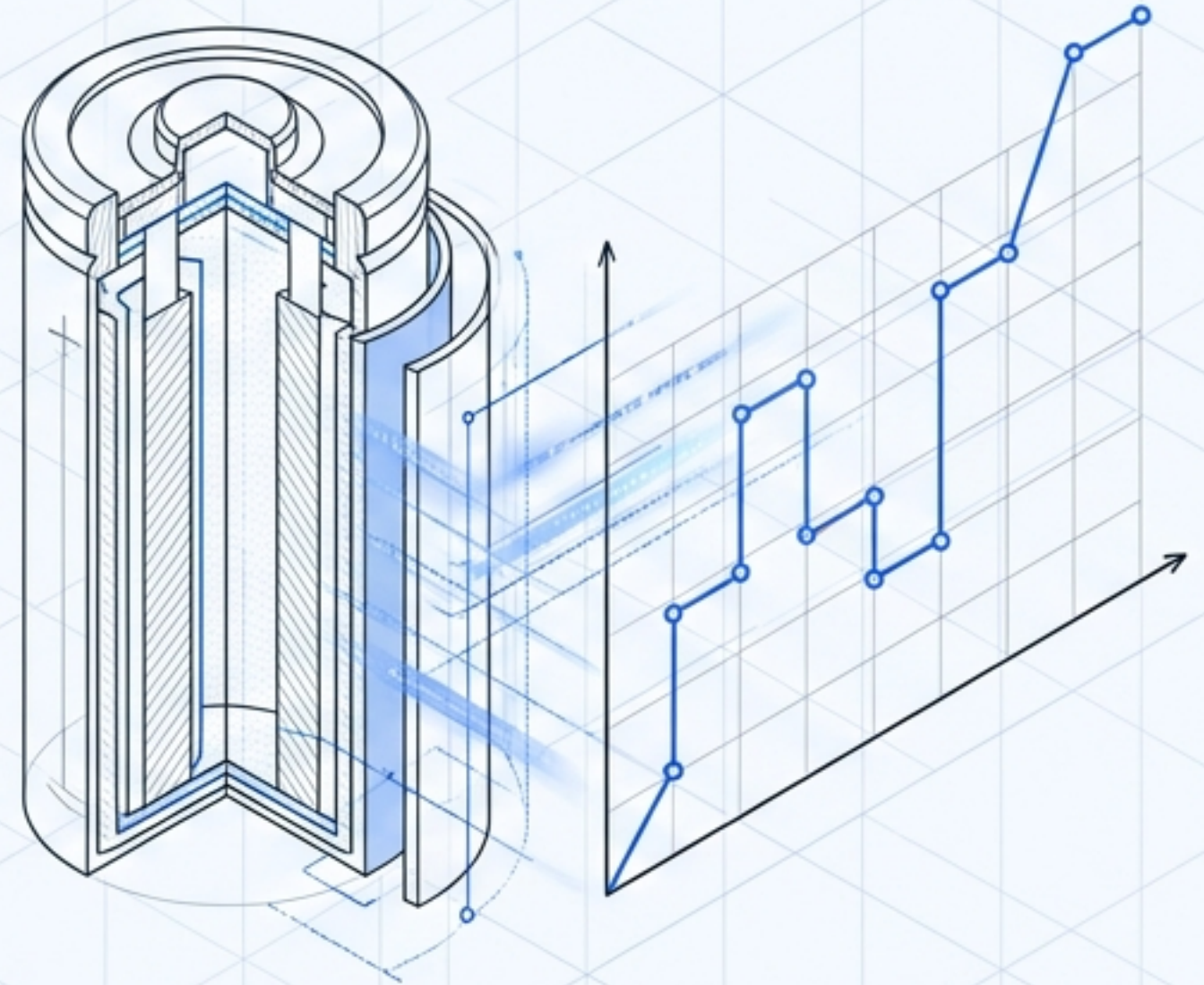


Battery SoC Modeling & Smart Charging

A `Python`-based simulation of rule-based control for residential Energy Storage Systems (BESS).



Core_Engine

Coulomb counting & linear degradation physics.

Strategy

Rule-based Time-of-Use (TOU) optimization.

Output

Deterministic 60-day performance simulation.

Bridging Energy Management & Battery Longevity

The Real-World Motivation



Grid Stress: Mitigating demand via peak shaving and load shifting.



Economics: Exploiting Time-of-Use (TOU) electricity tariffs to reduce cost.

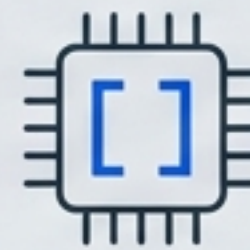


Longevity: Extending battery lifespan through constrained, optimized operation.

The Engineered Solution



Implement realistic **State of Charge (SoC) dynamics** and **efficiency losses**.



Develop rule-based smart charging logic.



Execute a **60-day simulation** to extract and analyze **performance metrics**.

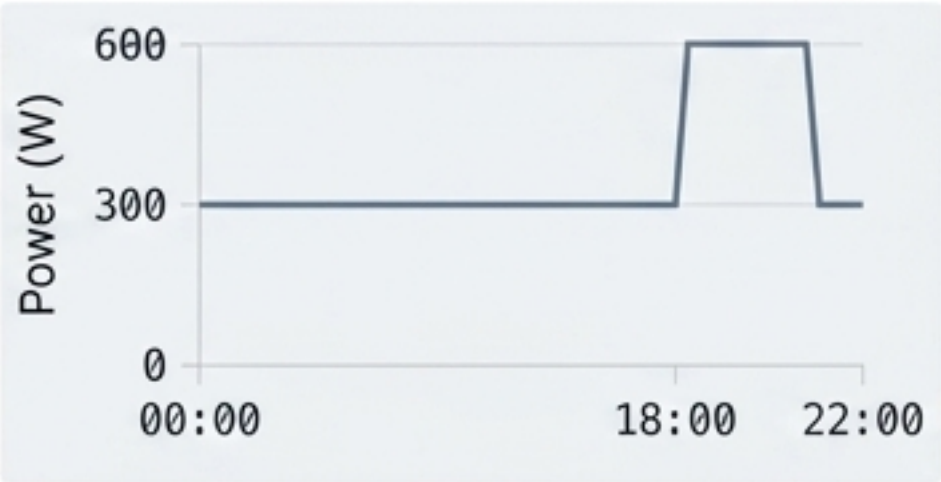
Defining the Simulation Environment

Battery Constraints

4 kWh Li-Ion

Usable Range:	20% - 90% (Optimized for health)
Efficiencies:	95% Charge / 95% Discharge
Power Limits:	Max 1000 W (In) / 1200 W (Out)
Degradation:	~5% loss per 100 equivalent cycles

Household Load Profile



Base Load:	300 W (Constant background draw)
Peak Load:	600 W (Evening spike: 18:00 - 22:00)

Simulation Parameters

Duration:	60 Days
Resolution:	1-Hour Time Steps
Tariff Structure:	Low-price window active 00:00 - 06:00

Translating Physics into Code

State of Charge (Coulomb Counting)

```

$$\Delta E = P_{\text{charge}} \times \Delta t \times \eta_{\text{charge}}$$

$$\Delta \text{SoC} = \Delta E / (\text{Capacity} \times \text{SoH})$$

$$\text{SoC} \leftarrow \min(\text{SoC} + \Delta \text{SoC}, 1.0)$$

```

Note: Accurately accounts for asymmetric charge/discharge efficiencies (η).

State of Health (Linear Degradation)

```

$$\text{SoH} = \max(0.7, 1 - (0.0005 \times \text{cycles}))$$

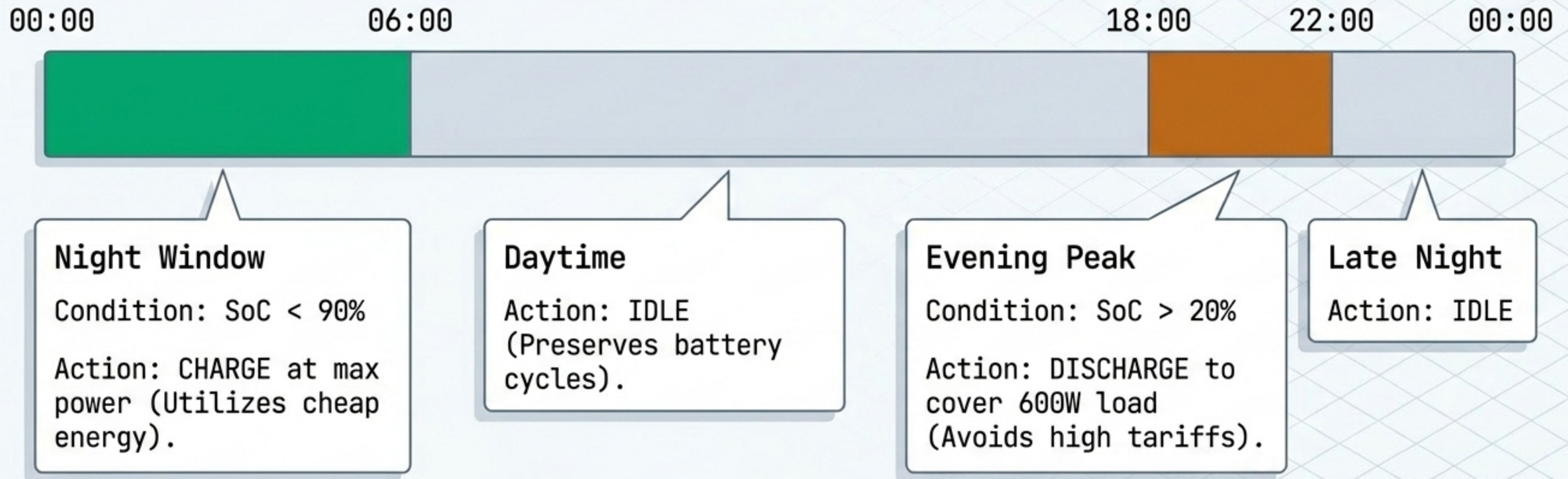

Constraint: Bottoms out at a minimum of 70% health.


```

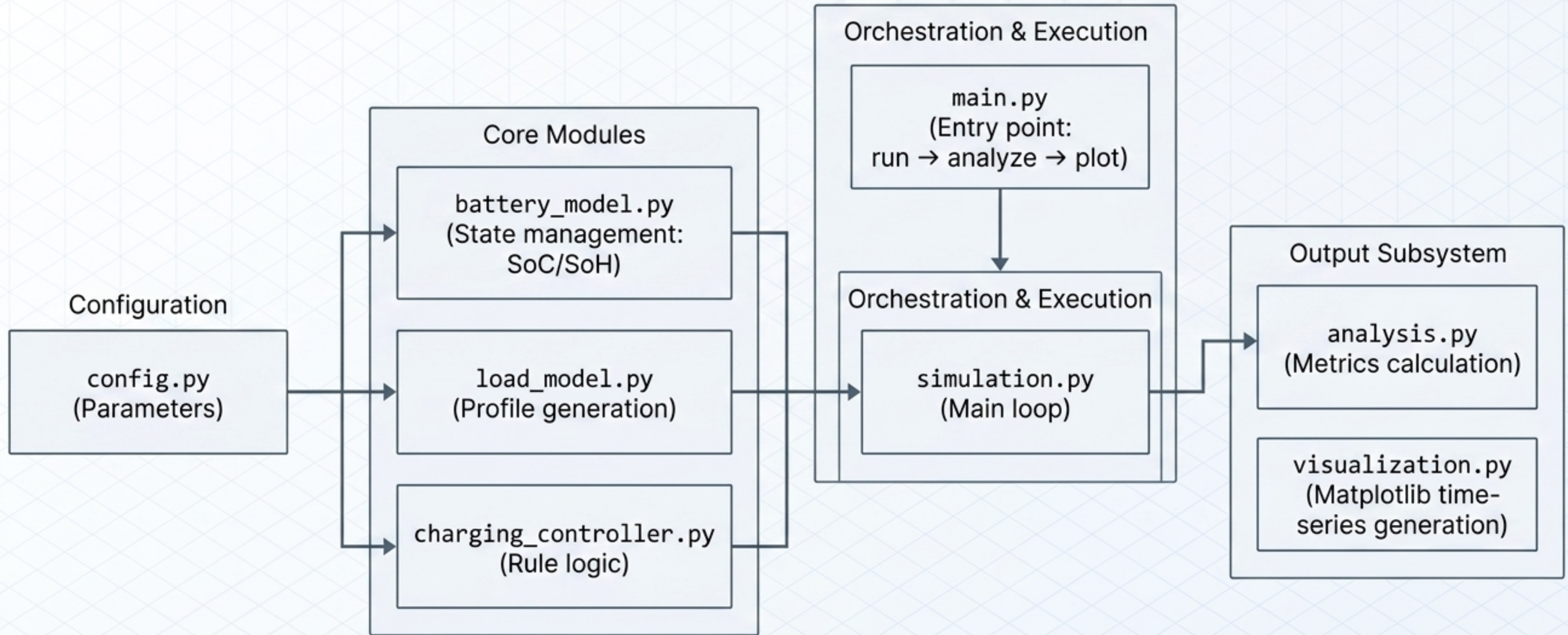
Note: Utilizes simplified daily cycle counting to model long-term capacity fade over the 60-day runtime.

Deterministic Logic for Daily Operations

Transparent, predictable logic that prioritizes cost reduction and safety limits without requiring complex machine learning overhead.



Modular Python Implementation



60-Day Performance & System Viability

47-49%

Average SoC

Balanced utilization

20.0%

Minimum SoC
Reached

Controller successfully honors
strict safety constraints

60-100

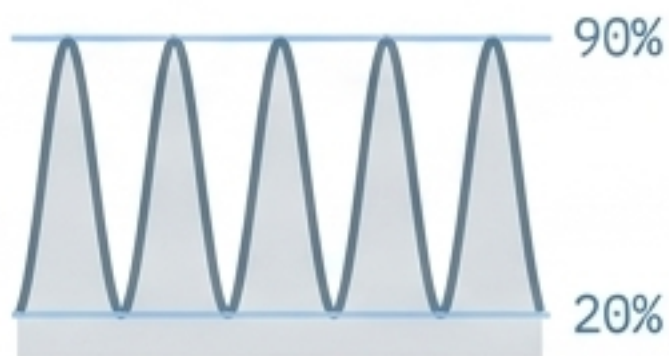
Equivalent Full
Cycles

97.0-97.5%

Final SoH

Realistic ~2.5% degradation
rate for light daily cycling

Simulation Observations



State of Charge Pattern:
Consistent daily cycling,
capping safely at 90%
during the night and
bottoming exactly at 20%
during evening peaks.

Simulation Observations



State of Health Pattern:
Predictable, gradual
linear decline validating
the physical models.

Educational Takeaways & Next Steps

Key Conclusions



Success of Simplicity: Basic, transparent logic effectively shifts load and reduces peak grid stress.



Safe Operation: Strict rule constraints successfully protected battery longevity without deep discharge.



Goal Achieved: Effectively demonstrated fundamental energy balance and efficiency losses in a residential BESS.

Future Scope (Beyond the Sandbox)



Integrate dynamic, real-world TOU tariff data APIs.



Add stochastic weather and load forecasting inputs.



Implement non-linear, temperature-dependent degradation models.